Curriculum Vitae of Katarzyna Wręczycka, PhD

katwre.github.io scholar.google.com/citations?user=katarzynawreczycka github.com/katwre

Profile

Computational biologist and machine-learning researcher with a PhD in Computational Molecular Biology and 9+ years of experience across academia and industry.

Work Experience

Computational Biologist & Machine Learning Scientist

02.2023 – present
Berlin, Germany
Zurich, Switzerland

Freelance | Freiberufler

Engaged in B2B contracts with pharmaceutical companies, start-ups and R&D consultancy companies. Key highlights include:

- **Anomaly detection for water safety monitoring** *01–06.2026*
  Developed an LSTM-based variational autoencoder in PyTorch to detect and forecast anomalous patterns of chemicals in high-frequency laser particle-counter measurements, and contributed to a Kafka-based streaming pipeline.
- **Menstrual blood biomarker discovery** *09–12.2025*
  Led a pilot project on classifying endometriosis, blood profiling, oncology biomarker discovery and immune cell dynamics using targeted RNA-seq and scRNA-seq from menstrual effluent via using generative modeling and transfer learning (scanpy, scvi-tools, scArches) and evaluated downstream classifiers (XGBoost) for endometriosis-associated signatures on AWS (EC2, S3).
- **Oncology biomarker discovery in healthcare** *05.2023–01.2025*
  - **Machine learning for drug development:** Applied positive unlabeled learning with random forests and SVM (with pulearn) and autoencoder (in PyTorch) to prioritize transcription factors for future clinical trials, integrated clinical-trial insights from GenomOncology database and public databases such as DepMap, TCGA, TEMPUS.
  - **Translational oncology signature discovery:** Collaborated closely with data scientists and experimental scientists in R&D and Clinical Development teams to identify tumor-associated antigens and gene signatures for patients with limited treatment options. Leveraged RNA-seq, WGS, external clinical trial databases, and performed survival analysis to support therapeutic decision-making.
- **Multimodal omics analysis in Alzheimer's disease** *03–05.2023*
  Investigated glia-to-neuron reprogramming by integrating gene expression and epigenomic data, including RNA-seq, ATAC-seq, ChIP-seq and DNA methylation data in reproducible Nextflow/nf-core workflows. Applied the hybrid CNN–Transformer Enformer model (Avsec et al., Nature Methods, 2021) to characterize regulatory enhancers and their potential target genes, and presented findings to wet-lab scientists to inform downstream experimental validation.

Computational Discovery Scientist

06.2026 – present
Zurich, Switzerland

Polaroid Therapeutics

- **Crohn's disease biomarker discovery:** Developing computational and machine-learning approaches for biomarker discovery (incl. mixed-effect regression, XGBoost) in Crohn's disease, integrating large-scale biobank data from the IBD Plexus of the Crohn's & Colitis Foundation, including blood-based proteomics (Olink Explore 3072).
- Processed and analysed on AWS (EC2, S3).

Bioinformatician | Research Assistant II

04.2025 – 06.2025
Zurich, Switzerland

ETH Zürich & Functional Genomics Center Zürich (FGCZ) of the University of Zurich (UZH)
Employed by the Prof. von Meyenn Lab (40% position), Prof. Bar-Nur Lab (40%) and FGCZ/UZH (20%)

- **Integrative omics analyses across multiple projects:** Performed gene-regulatory analyses for muscle aging with natural compounds (RNA-seq), germline chimerism (snRNA-seq), yo-yo effect in adipocytes (CUT&TAG), and assembly of Y chromosome of *Mus minutoides* (WGS).
- **Cross-disciplinary collaboration** with wet-lab scientists to interpret analysis and experiments.

- **Reproducible NGS workflows on HPC:** Maintained and operated end-to-end Nextflow pipelines (RNA-seq, CUT&TAG) via Nextflow Tower on HPC (SLURM) at ETH Zürich, using Docker/Singularity and Spack-managed environments.
- **ShinyProxy for snRNA-seq:** Deployed interactive R Shiny apps via ShinyProxy for snRNA-seq datasets (shiny-public.fgcz.uzh.ch; snRNA-seq: MHUO, Mouse epiAT memory, Human AT memory).

PhD Student in Computational Molecular Biology — *10.2015 – 11.2021*

Berlin Institute for Medical Systems Biology (BIMSB) at the Max Delbrück Center for Molecular Medicine (MDC) & Humboldt University of Berlin. — *Berlin, Germany*

- Led and contributed to research projects applying statistical and ML methods, resulting in 4 **publications with 420+ citations**.
- **Genomic artifact detection**: Investigated high-density transcription factor binding sites (HOT regions) and identified spurious ChIP-seq peaks linked to RNA structures called DNA-RNA hybrids (R-loops), and hypo-methylated regions using elastic net regression and PCA. Integrated data from publicly available resources, including ENCODE and Roadmap Epigenomics, as well as experimental datasets from ChIP-seq, ATAC-seq, Bisulfite-seq, and DRIP-seq/RDIP-seq studies [2].
- **DNA methylation biomarkers for acute coronary syndrome discovery**: Studied Bisulfite-seq data from patient-derived heart tissue and plasma, uncovering gene and transcription factor associations. My contribution included preprocessing raw Bisulfite-seq data, calling differentially methylated regions, associating them with genes, and transcription factors in cardiovascular diseases [1].
- **Snakemake pipeline for Bisulfite-seq**: A scalable pipeline using TrimGalore!, BWA-meth/Bismark, samblaster, MethylKit/methylDackel, and genomation R packages, published as part of the PiGX: Pipelines in Genomics project [3].
- First co-author of a peer-reviewed **review on statistical methods for analyzing Bisulfite-seq data** [4].
- **Developed the R package motifActivity** for transcription factor network reconstruction from sequencing data using rank-based estimation linear models.
- **Multi-omics analysis of high-risk pediatric neuroblastoma**: Identified DNA methylation biomarkers and transcription factor networks specific to high-risk cases from solid tumors and cell-free DNA in urine. Applied machine learning methods (incl. elastic net, ridge regression, XGBoost) and integrated patient-derived samples from Bisulfite-seq, RNA-seq, and ChIP-seq datasets in collaboration with Dr. med. Johannes H. Schulte from Charité Hospital in Berlin.

Scholarships

Received a €7,000 scholarship from the Institute of Computer Science, Polish Academy of Sciences (IPI PAN), co-funded by the European Union funds, for predoctoral research at the MDC-BIMSB. Also co-developed genomation, a Bioconductor tool for genomic data visualization and annotation. — *03 – 09.2015, Berlin, Germany*

Education

Ph.D. in Computational Molecular Biology — *10.2015 – 05.2021, Berlin, Germany*

Humboldt University of Berlin
Advisors: Dr. Altuna Akalin and Prof. Dr. Uwe Ohler
Thesis: "Integrative analysis on the effect of DNA methylation on gene regulation in tumor and healthy tissues" (Link to PDF).

M.Sc. in Bioinformatics — *10.2012 – 06.2014, Warsaw, Poland*

University of Warsaw, Department of Mathematics, Informatics and Mechanics
Advisor: Prof. Dr hab. Anna Gambin
Thesis: "Genome-wide analysis of noncoding copy-number variations in human genome".

B.Sc. in Bioinformatics — *10.2010 – 06.2012, Warsaw, Poland*

University of Warsaw, Department of Mathematics, Informatics and Mechanics
Advisor: Dr hab. Bartosz Wilczynski
Thesis: "Potential regulatory region discovery method implementation".

Publications

[1] R. R. C. Cuadrat, A. Kratzer, H. G. Arnal, R. A. C., K. Wreczycka, A. Blume, V. Ebenal, T. Mauno, B. Osberg, M. Moobed, J. Hartung, C. Seppelt, D. Meteva, A. Haghikia, D. Leistner, U. Landmesser and A. Akalin, "Cardiovascular disease biomarkers derived from circulating cell-free dna methylation," NAR Genomics and Bioinformatics, 2023. DOI: 10.1093/nargab/lqad061.

[2] K. Wreczycka, V. Franke, B. Uyar, R. Wurmus, S. Bulut, B. Tursun and A. Akalin, "HOT or not: examining the basis of high-occupancy target regions," Nucleic Acids Research, vol. 47, no. 11, pp. 5735–5745, May 2019. DOI: 10.1093/nar/gkz460.

[3] R. Wurmus, B. Uyar, B. Osberg, V. Franke, A. Gosdschan, K. Wreczycka, J. Ronen and A. Akalin, "PiGx: reproducible genomics analysis pipelines with GNU Guix," GigaScience, vol. 7, no. 12, Oct. 2018, giy123. DOI: 10.1093/gigascience/giy123.

[4] K. Wreczycka, A. Gosdschan, D. Yusuf, B. Grüning, Y. Assenov and A. Akalin, "Strategies for analyzing bisulfite sequencing data," Journal of Biotechnology, vol. 261, pp. 105–115, 2017, Bioinformatics Solutions for Big Data Analysis in Life Sciences presented by the German Network for Bioinformatics Infrastructure. DOI: 10.1016/j.jbiotec.2017.08.007.

Conferences & workshops

Talks & Conference Presentations

- **Speaker**, "Assessing prognostic and diagnostic features of neuroblastoma methylome", DKTK – Young Academics Meeting, 6 Sep. 2019, Berlin, Germany
- **Poster**, "Assessing prognostic and diagnostic features of high-dimensional (epi)genomics data for neuroblastoma", RECOMB, 5–8 May 2018, Washington, D.C., USA
- **Speaker**, "HOT or not: Examining the basis of high-occupancy target regions", CRG-BIMSB MDC PhD Retreat, 15–18 May 2017, Girona, Spain
- **Poster presentation**, "HOT or not: redefining the origin of high-occupancy target regions", EMBL Conference "From Functional Genomics to Systems Biology", 12–15 Nov. 2016, Heidelberg, Germany
- **Speaker**, "Genomation: an R package to summarize, annotate and visualize genomic intervals", Bioinformatics Social Meetings Berlin, 16 Sep. 2016, Berlin, Germany
- **Poster presentation**, "Genomation: an R package to summarize, annotate and visualize genomic intervals", Max-Planck-Gesellschaft Otto Warburg Summer School, 6–11 Sep. 2015, Berlin, Germany

Workshops & Training (selected)

- Parallelisation with OpenMP, Swiss Institute of Bioinformatics workshop, 11 May. 2025, Switzerland (, GPU computing)
- Ensuring More Accurate, Generalisable, and Interpretable Machine Learning Models for Bioinformatics, Swiss Institute of Bioinformatics workshop, 17 Nov. 2025, Switzerland (, XGBoost Bayesian optimization)
- Machine Learning Zoomcamp, Data Talks Oct 2025 - Jan 2026 ( )
  - Immune cell classifier: Developed and trained a CNN-based image classifier for immune cell detection in H&E-stained histopathological images using Xception with an MLP head; dockerized; deployed on AWS Lambda for scalable inference .
  - Diffusion models for immune cell images: Developed a UNet-based diffusion model with linear noise scheduling for H&E-stained immune cell image inpainting, with an interactive Streamlit frontend for model exploration, deployed on AWS EC2 .
  - Gene type prediction from DNA sequence: Designed and trained a transformer encoder for genomic sequence classification; implemented ONNX Runtime for inference and deployed a FastAPI and BentoML service on AWS Kubernetes .
- Using Large Language Models for Biodata Exploration: From Theory to Practice, Swiss Institute of Bioinformatics workshop, 30 Sept. 2025, Zurich, Switzerland (LLM-based AI assistants)
- PyData Berlin Conference and Workshops, 1–3 Sep. 2025, Berlin, Germany (State-Space Models with PyMC)
- Federated Learning in Bioinformatics, Swiss Institute of Bioinformatics workshop, 29 Aug. 2025, Lugano, Switzerland (, VAE to mitigate batch effects in scRNA-seq using federated learning)
- EMBL-EBI Cancer Genomics course, 20–23 June 2017, Hinxton, United Kingdom